



WIRESHARK PROTOCOL & PACKET ANALYSIS

AQUAMAN EDITION

Deep-Sea Network Inspection, OSI Dissection & Display Filter Master Reference

1. TOOL OVERVIEW & GENERAL OPERATIONS

1.1 Use Cases & Core Architecture

- **Primary Functions:** Network troubleshooting, bottleneck detection, security anomaly hunting (malicious traffic, abnormal ports, rogue hosts), and protocol analysis [cite: 1, 4, 5].
- **Important Guardrail:** Wireshark is **NOT** an Intrusion Detection System (IDS) [cite: 1, 4, 5]. It is a passive packet analyzer (packet sniffer) [cite: 1, 4, 5]. It reads data without modifying or altering packets [cite: 1, 4, 5]. Finding anomalies relies entirely on analyst investigation skills [cite: 1, 4, 5].

1.2 GUI & Data Layout

Panel Name	Function & Content
Packet List Pane	Summary table of captured traffic (Packet #, Time, Source/Destination IPs, Protocol, Length, Info) [cite: 1, 4, 5].
Packet Details Panel	Hierarchical protocol breakdown of the selected packet structured by OSI layers [cite: 1, 4, 5].
Packet Bytes Pane	Raw data output in Hexadecimal and decoded ASCII formats [cite: 1, 4, 5].

1.3 Key Interface Elements

- **Toolbar:** Shortcuts for start/stop sniffing, filtering, sorting, merging, and exporting [cite: 1, 4, 5].
- **Display Filter Bar:** Core query bar used to filter displayed packets [cite: 1, 4, 5].
- **Status Bar:** Bottom bar displaying active profile, total captured packets, displayed packet counts, and Expert Information alerts [cite: 1, 4, 5].

1.4 Capture File Metadata & Properties

- **Path:** Statistics -> Capture File Properties [cite: 1, 4, 5].
- **File Hash:** SHA256, SHA1, and MD5 cryptographic integrity signatures [cite: 1, 4, 5].
- **File Comments:** Analyst notes embedded inside .pcapng files (frequently contains CTF flags) [cite: 1, 4, 5].
- **Statistics:** Exact packet counts, average packet size, capture duration, and hardware interface details [cite: 1, 4, 5].

1.5 File Operations

- **Merge PCAPs:** Combines multiple capture files chronologically (File -> Merge) [cite: 1, 4, 5].
- **Command Line Count (capinfos):** capinfos -c file.pcapng prints the exact packet count without launching the GUI [cite: 1, 4, 5].

3. TCP PROTOCOL & PERFORMANCE METRICS

3.1 TCP Header Structure (20-60 Bytes)

```

 0                               1
 2                               3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6
 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                               Source Port                       |
|                               Destination                       |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                               Sequence Number                 |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                               Acknowledgment Number           |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
| Data | Res | Control Flags |                               Window
Size |                               |
|Offset|                               |C|E|U|A|P|R|S|
F|                               |                               |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                               Checksum                          |
|                               Urgent                            |
|                               Pointer                           |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                               Options (0 to 40
Bytes)                               |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

3.2 Core Networking Metrics Reference

Metric	Layer	Core Function
MTU	Layer 3	Max packet size passing interface without fragmentation (Standard: 1500 Bytes) [cite: 1, 4, 5].
MSS	Layer 4	Max payload: $MSS = MTU - (IP\ Header + TCP\ Header) = 1460\ B$ [cite: 1, 4, 5].
SEQ Number	Layer 4	Tracks byte position in stream to guarantee reordering/delivery [cite: 1, 4, 5].
Window Size	Layer 4	Maximum unacknowledged byte volume host can buffer (Flow Control) [cite: 1, 4, 5].
RTT	Latency	Total transit time (ms) for segment to reach receiver + ACK return [cite: 1, 4, 5].
Bytes in Flight	Wireshark	Total unacknowledged data payload traveling through pipeline [cite: 1, 4, 5].

Packet Burst Formula

$Number\ of\ Packets\ in\ Flight = Window\ Size / MSS$ [cite: 1, 4, 5]

2. OSI LAYER PACKET DISSECTION

Packet dissection breaks down raw bytes into structured protocol fields according to the OSI model [cite: 1, 4, 5].

```
+-----+
+-----+
| Layer 1: Frame / Physical      (Arrival Time, Frame
|                               Length)          |
+-----+
+-----+
| Layer 2: Data Link             (Source/Destination MAC
|                               Addresses)        |
+-----+
+-----+
| Layer 3: Network               (Source/Destination IP
|                               Addresses, TTL)|
+-----+
+-----+
| Layer 4: Transport             (TCP/UDP Ports, SEQ/ACK,
|                               Window Size)|
+-----+
+-----+
| Layer 5/7: Application         (HTTP, DNS, FTP, SMB,
|                               ETag, Markup)    |
+-----+
+-----+
```

2.1 Layer-by-Layer Dissection Guide

-  **Frame (OSI Layer 1 - Physical):** Contains hardware frame length, interface IDs, epoch timing, and exact Arrival Time (Timestamp) [cite: 1, 4, 5].
-  **Data Link (OSI Layer 2):** Ethernet headers containing Source and Destination MAC Addresses [cite: 1, 4, 5].
-  **Network (OSI Layer 3):** IPv4/IPv6 headers containing Source/Destination IP Addresses and Time to Live (TTL) [cite: 1, 4, 5].
-  **Transport (OSI Layer 4):** TCP/UDP protocols, Source/Destination Ports, Sequence Numbers (SEQ), Acknowledgment Numbers (ACK), Window Size, and TCP Payload Size [cite: 1, 4, 5].
-  **Application Protocol (OSI Layer 5-7):** Protocol-specific metadata (e.g., HTTP ETag, Request Methods, User-Agent, response status codes) [cite: 1, 4, 5].
-  **Application Data:** Reassembled raw content payload (e.g., HTML/XML markup code, images, files) [cite: 1, 4, 5].

4. PACKET NAVIGATION & ANALYSIS TOOLS

4.1 Navigation Actions

- **Go to Packet (Ctrl + G):** Instantly jumps to a specific packet number [cite: 1, 4, 5].
- **Find Packet (Ctrl + F):** Searches packet data by String, Hex, Regex, or Display Filter [cite: 1, 4, 5].
-  **Critical Rule:** Select the correct pane to search! Searching in Packet Details won't locate strings existing only in Packet Bytes [cite: 1, 4, 5].

4.2 Analyst Annotation & Extraction

Feature	Menu Path	Retention	Description
Mark Packet	Right-click -> Mark/Unmark (Ctrl + M)	Temporary	Turns packet black for session tracking [cite: 1, 4, 5].
Packet Comment	Right-click -> Packet Comment	Permanent	Attaches notes inside .pcapng structure [cite: 1, 4, 5].
Export Packets	File -> Export Specified Packets	Permanent	Saves targeted subset to new file [cite: 1, 4, 5].
Export Objects	File -> Export Objects -> [HTTP/SMB/etc.]	Permanent	Extracts transmitted files (images, EXEs, PDFs) [cite: 1, 4, 5].

4.3 Expert Information Diagnostic Levels

Access via status bar bottom-left indicator or Analyze -> Expert Information [cite: 1, 4, 5]:

-  **Error (Red)** Malformed packets or protocol crashes [cite: 1, 4, 5].
-  **Warn (Yellow)** Unusual error codes or connection resets [cite: 1, 4, 5].
-  **Note (Cyan)** Notable application events (e.g., HTTP 404 response) [cite: 1, 4, 5].
-  **Chat (Blue)** Routine protocol flow information [cite: 1, 4, 5].

5. PACKET FILTERING MASTER LIST

🔍 **The Golden Rule of Wireshark GUI:** "If you can click on it, you can filter and copy it." [cite: 1, 4, 5]

5.1 Context Menu Filtering Options

- **Apply as Filter:** Instantly filters selected value (Selected, Not Selected) [cite: 1, 4, 5].
- **Prepare as Filter:** Writes syntax in filter bar without executing immediately (allows building compound queries with and/or) [cite: 1, 4, 5].
- **Conversation Filter:** Isolates the complete 5-tuple stream between two endpoints (Source/Dest IP, Source/Dest Port, Protocol) [cite: 1, 4, 5].
- **Colourise Conversation:** Temporarily colors connected streams without hiding other capture traffic [cite: 1, 4, 5].
- **Apply as Column:** Converts a packet detail field into a top-level table column in Packet List Pane [cite: 1, 4, 5].
- **Follow Stream:** Reconstructs raw application-level data stream (TCP/UDP/HTTP). Client data appears in Red, Server response appears in Blue [cite: 1, 4, 5].

5.2 Essential Display Filter Cheat Sheet

```
# Protocol Filters
http
dns
arp
dhcp
icmp
ftp

# Port Filters
tcp.port == 80           # Filter all HTTP traffic
udp.port == 53           # Filter all DNS traffic
tcp.port == 443          # Filter all HTTPS traffic
tcp.port == 21           # Filter FTP control commands

# Host & IP Filters
ip.addr == 192.168.1.1   # Show traffic where IP is
                        # Source OR Destination
ip.src == 10.0.0.5
# Show traffic originating ONLY from this IP
ip.dst == 10.0.0.1
# Show traffic targeted ONLY to this IP

# Logical Operators & Compound Queries
# Show HTTP traffic from a specific IP
ip.addr == 192.168.1.15 and http

# Show traffic excluding local net broadcast
not ip.addr == 192.168.1.255

# Target specific TCP flags during handshake
tcp.flags.syn == 1 and tcp.flags.ack == 0

# Search for specific string inside HTTP responses
http.response.line contains "200 OK"
```